Design of Loss function

anonymous

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1 Empirical Analysis

To empirically evaluate the performance of the proposed loss functions, we design a comprehensive asset pricing experiment. This section details the data, the construction of predictor variables, the backtesting procedure, and the evaluation metrics used to compare the models.

1.1 Data

The goal is to acquire monthly stock data from 1990 to 2025 to predict one-month-ahead returns.

1. WRDS Account Registration:

- Navigate to the WRDS website: https://wrds-www.wharton.upenn.edu/.
- Register for an account using your official XJTLU university email. Be sure to select ”XJTLU” from the institutional drop-down menu.
- Account confirmation typically takes 1-2 business days.
- Important: You may receive an email from the university library asking for the research rationale. Please reply promptly, stating it is for academic research, and CC me on your response.

2. Data Extraction Procedure:

- Once logged in, find the Center for Research in Security Prices, LLC (CRSP) dataset under the “Subscriptions” heading on the homepage.
- Navigate through the following menus: “CRSP” → “Stock / Security Files” → “Monthly Stock File”.

3. Query Form Configuration:

What we need to do is to use the features extracted from the monthly data to predict the stock return one month ahead. Works in finance literature typically predict returns at monthly or annual interval, such as Han (2021); Gu et al. (2020, 2021).

When you enter the “CRSP Monthly Stock” page, you may see a lot of options. Here is some suggestions for selecting data.

- “Step 1: Choose your date range”: Date Range: To avoid download errors from excessively large files, download the data in smaller increments (e.g., 1-2 years at a time). The complete required range is 1990 to 2025.

- “Step 2: Apply your company codes”: Select the option to “Search the entire database”

- “Step 3: Choose query variables”: Select the following essential variables: “(ret) Returns”, “(vol) Volume”, “(shrout) Number of Shares Outstanding”. Do not “select all” to keep the file size manageable. You can review variable definitions by clicking the question mark icon next to each.

- “Step 4: Select query output”: Select the output format as CSV (.csv) and choose Uncompressed. You may select your preferred date format.

- “Step 5: Download”: Click ”Submit Form” and wait for the data file to be generated. Download the file and repeat the process until you have collected data for the entire 1990-2025 period.

Overall, we need at least 35 years data for empirical analysis. The data can cover the time periods from 1990 to 2025, which cover most of the well-known economic booming and recession periods.

1.2 Model Input and Output

The objective of our models is to predict the one-month-ahead stock return, $r_{i,t+1}$, for each stock i. This aligns with the standard asset pricing literature on return predictability Gu et al. (2020); Han (2021). The predictive relationship is modeled as:

$$r_{i,t+1} = f(\mathbf{X}_{i,t}) + \epsilon_{i,t+1}$$

where $f(\cdot)$ is the predictive model and $\mathbf{X}_{i,t}$ is the set of predictor variables (features) for stock i at time t. We construct three distinct sets of features, which are tested independently to ensure a robust evaluation.

Feature Set 1 ($\mathbf{X}^1_{i,t}$): We first define the input set $\mathbf{X}_{i,t}$. We consider three sets of input variables. Following Medhat and Schmeling (2021), we define our first set of variables, $\mathbf{X}^1_{i,t}$, as:

$$\begin{aligned}
\mathrm{cr}_{i,m} &= \prod_{j=t-m}^{t-1} (r_{i,j}+1) - 1,\ \ m = 1, 3, 6, 9, 12 \\
r_{i,t} &= \frac{p_{i,t}}{p_{i,t-1}} - 1 \\
\mathrm{co}_{i,m} &= \sum_{j=t-m}^{t-1} \left(\mathrm{to}_{i,j}\right),\ \ m = 1, 3, 6, 9, 12 \\
\mathrm{to}_{i,t} &= \frac{\mathrm{vol}_{i,t}}{\mathrm{os}_{i,t}}
\end{aligned} \tag{1}$$

For stock i at time t, these variables are defined as follows: $r_{i,t}$ represents the simple return, whilst $\mathrm{vol}_{i,t}$ denotes the total trading volume from $t-1$ to t. The variable $\mathrm{os}_{i,t}$ represents the outstanding shares, and $\mathrm{to}_{i,t}$ denotes the turnover. Additionally, $\mathrm{cr}_{i,m}$ and $\mathrm{co}_{i,m}$ represent the cumulative return and cumulative turnover, respectively, calculated over the past m months, where m takes values of 1, 3, 6, 9, and 12 months.

Feature Set 2 ($\mathbf{X}^2_{i,t}$): Drawing from Daniel and Moskowitz (2016); Han (2021), our second set of variables $\mathbf{X}^2_{i,t}$ is defined as:

$$\begin{aligned}
\mathrm{cr}^{-}_{i,m} &= \prod_{j=t-m}^{t-2} (r_{i,j}+1) - 1,\ \ m = 3, 6, 9, 12 \\
\mathrm{Ncr}_{i,m} &= \frac{\mathrm{cr}^{-}_{i,m} - \mathrm{Avg}\ \mathrm{cr}^{-}_{m}}{\mathrm{Std}\ \mathrm{cr}^{-}_{m}}
\end{aligned} \tag{2}$$

where $\mathbf{X}^2_{i,t}$ consists of the normalised cumulative return ($\mathrm{Ncr}_{i,m}$) calculated over m months, where m takes values of 3, 6, 9, and 12. This measure excludes the most recent month's return. $\mathrm{Avg}\ \mathrm{cr}^{-}_{m}$ represents the cross-sectional average of $\mathrm{cr}^{-}_{i,m}$, whilst $\mathrm{Std}\ \mathrm{cr}^{-}_{m}$ denotes its cross-sectional standard deviation.

Feature Set 3 ($\mathbf{X}^3_{i,t}$): Our third set of input variables, $\mathbf{X}^3_{i,t}$, follows Daniel and Moskowitz (2016); Han (2021) and incorporates normalised monthly returns ($\mathrm{nr}_{i,j}$) for the preceding 12 months. These returns are defined as:

$$\mathrm{nr}_{i,j} = \frac{r_{i,j} - \mathrm{Avg}\ r_j}{\mathrm{Std}\ r_{i,j}},\ \ j = t-m, m = 1, \ldots, 12 \tag{3}$$

where Avg r_j represents the cross-sectional average of raw returns, and Std $r_{i,j}$ denotes their cross-sectional standard deviation. The normalisation is computed for each month j, ranging from $t-1$ to $t-12$. This set consists of 12 features. Table 1 summarizes the feature sets.

Table 1: This table summarises the variables in three input sets.

Input Sets	dimensions	past months	Input Variables
Set1 $\mathbf{X}^1_{i,t}$	5×2=10	m=1,3,6,9,12	$\mathrm{cr}_{i,m}$, $\mathrm{co}_{i,m}$
Set2 $\mathbf{X}^2_{i,t}$	4×2=8	m=3,6,9,12	$\mathrm{Ncr}_{i,m}$, Avg cr^-_m
Set3 $\mathbf{X}^3_{i,t}$	12×2=24	j=t-m, m=1,2,$\cdots$,12	$\mathrm{nr}_{i,j}$, Avg r_j

1.3 Training Methodology and Portfolio Strategy

We employ a rolling-window backtesting procedure to simulate a realistic trading environment and mitigate look-ahead bias. The process is as follows:

1. **Initial Training:** The first model is trained on an initial 5-year data window (e.g., Jan 1990 - Dec 1994).

2. **Prediction Period:** The trained model is used to generate one-month-ahead return predictions for the subsequent 6 months (e.g., Jan 1995 - Jun 1995).

3. **Re-training:** The training window is rolled forward by 6 months (e.g., the new window becomes Jul 1990 - Jun 1995). A new model is trained on this updated dataset.

This cycle of predicting for 6 months and re-training is repeated until the end of our sample period.

For each month in the prediction period, we form a long-short portfolio based on the model's return forecasts. All stocks are sorted in descending order of their predicted returns. We take a long position in the top decile (top 10%) and a short position in the bottom decile (bottom 10%). The performance of these equal-weighted portfolios is then tracked over time.

1.4 Models

1.4.1 Basic Models

We evaluate a suite of nine models, all based on a feed-forward neural network (NN) architecture but differing in their loss function.

- **Baseline Models:** We establish a benchmark using standard loss functions: Mean Squared Error (NN_{Mean}) and Median Squared Error (NN_{Median}). The hyperparameters (e.g., number of layers, neurons) for a simple NN are tuned via grid search for the NN_{Mean} model, and this fixed architecture is then used for all subsequent models to ensure a fair comparison of the loss functions.

- **MADL-based Models:** We implement models using the Mean Absolute Deviation Loss (NN_{MADL}) and its generalized version (NN_{GMADL}), following the work of Michańków et al. (2024b,a).

- **Proposed Loss Function Models:** We implement NNs with each of the proposed loss functions in your document (but before implementation, you shall clearly understand the rationale of these loss function in the behind).

1.5 Performance Evaluation Metrics

To provide a thorough comparison, we assess both the predictive accuracy and the economic value of the models using the following metrics:

- **Predictive Accuracy:** We calculate the out-of-sample Root Mean Squared Error (RMSE) for the entire test period. We also plot the rolling 12-month RMSE to analyze how predictive performance evolves over time.

- **Portfolio Performance:** We compute and plot the cumulative returns of the long, short, and long-short portfolios.

- **Risk-Adjusted Returns:** We calculate the overall Sharpe Ratio for the long-short portfolio over the entire test period. Additionally, we plot the rolling 12-month Sharpe Ratio to assess the stability of risk-adjusted performance.

Research To-Do List

Phase 1: Data Acquisition

1. **Register for WRDS Access:**

    - Go to the WRDS website: https://wrds-www.wharton.upenn.edu/.

- Register for an account using your official university email (`@xjtlu.edu.cn`). Select "Xi'an Jiaotong-Liverpool University" from the institution list.
- Wait for approval (1-2 business days). You may receive an email from the university library asking for the research rationale. Reply promptly, explaining the project is for academic research, and CC me on the email.

2. **Download CRSP Monthly Data:**

   - Once logged in, navigate to CRSP under "Subscriptions".
   - Select "Stock / Security Files" → "Monthly Stock File".
   - **Step 1 (Date Range):** Download the data in manageable chunks (e.g., 5-10 years at a time) to avoid timeouts. The full range is **Jan 1990 to Dec 2025**.
   - **Step 2 (Company Codes):** Select "Search the entire database".
   - **Step 3 (Query Variables):** Select at a minimum: `PERMNO`, `DATE`, `RET` (Returns), `VOL` (Volume), and `SHROUT` (Shares Outstanding). Familiarize yourself with the definitions by clicking the question mark next to each variable.
   - **Step 4 (Query Output):** Choose output format as **CSV (.csv)** and compression as **Uncompressed**.
   - Submit the query and download the data files. Consolidate them into a single master data file.

Phase 2: Data Preprocessing & Feature Engineering

1. **Clean Data:** Handle missing values for `RET`, `VOL`, and `SHROUT`.
2. **Calculate Base Variables:** Compute simple returns ($r_{i,t}$) and turnover ($\text{to}_{i,t}$).
3. **Engineer Feature Sets:** Write scripts to generate the three feature sets ($\mathbf{X}^1, \mathbf{X}^2, \mathbf{X}^3$) for every stock-month observation as defined in the methodology. This will involve significant data manipulation (e.g., rolling products, cross-sectional normalization).
4. **Define Target Variable:** The target variable is the one-month-ahead return, $r_{i,t+1}$.

Phase 3: Model Implementation & Training

1. **Setup Baseline NN:**
    - Implement a standard feed-forward neural network in Python (using TensorFlow or PyTorch).
    - Perform a grid search for the optimal hyperparameters (e.g., layers, neurons, learning rate, activation function) using the NN_{Mean} model and one of the feature sets (e.g., $\mathbf{X}^1$).
    - **Crucially, fix this architecture.** This exact structure will be used for all other models to ensure we are comparing the loss functions, not the architectures.

2. **Implement All (or most) Loss Functions:**
    - Code the custom loss functions for all proposed ones.

3. **Implement Backtesting Loop:**
    - Create the rolling-window training script as described in Section 1.3 (5-year train, 6-month test, 6-month slide).
    - This loop will iterate through time, re-training each of the nine models at every step and storing the out-of-sample predictions.

Phase 4: Analysis & Results Generation

1. **Run Full Backtest:** Execute the backtesting script for all nine models on all three feature sets. This will be computationally intensive.

2. **Calculate Performance Metrics:** For each model and feature set combination:
    - Calculate and tabulate the overall out-of-sample RMSE.
    - Generate data for the rolling 12-month RMSE plots.
    - Calculate the monthly returns for the long, short, and long-short decile portfolios.
    - Generate data for the cumulative return plots.
    - Calculate and tabulate the overall Sharpe Ratio for the long-short portfolio.
    - Generate data for the rolling 12-month Sharpe Ratio plots.

3. **Visualize Results:** Create high-quality plots for the paper (cumulative returns, rolling RMSE, rolling Sharpe Ratios).

4. **Summarize Findings:** Create tables comparing the key metrics (overall RMSE, overall Sharpe Ratio) across all models and feature sets.

2 Appendix

References

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